Lecture 11

Goals: ① Review Bell States.
② No-signaling Condition.
③ Application of an Entangled State: PtI, Quantum Dense Coding
④ Holevo Bound.
⑤ Next Lecture: Quantum Teleportation.

Spooky Action at a Distance Property.

$$\begin{cases} |B_0\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \\ |B_1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |B_2\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |B_3\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \end{cases}$$

$$\left.\begin{aligned} (I\otimes I)|B_0\rangle &= (+1)|B_0\rangle \\ (I\otimes X)|B_0\rangle &= (+1)|B_2\rangle \\ (I\otimes Y)|B_0\rangle &= (-i)|B_3\rangle \\ (I\otimes Z)|B_0\rangle &= (-1)|B_1\rangle \end{aligned}\right\}$$

you can create 4 orthogonal states by operating on 1 qubit.

This almost instantaneous change violates relativity.

This is resolved by the No-Signaling Condition.

Suppose Alice applies an X gate to her qubit

$$(I\otimes X)|B_0\rangle = (I\otimes X)\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |B_2\rangle$$

Does Bob have any way of telling if Alice changed the state?

$$\forall \text{Bell State}, \begin{cases} \langle B_i|X\otimes I|B_i\rangle = 0 \\ \langle B_i|Y\otimes I|B_i\rangle = 0 \\ \langle B_i|Z\otimes I|B_i\rangle = 0 \end{cases}$$

Eigenvalues of X, Y, Z are ± 1. $\Rightarrow P_{+1} = P_{-1} = \frac{1}{2}$.

If one measure joint property $\underbrace{Z \otimes Z}_{\downarrow} \rightarrow \begin{matrix} -1 & \text{in } |B_0\rangle \\ +1 & \text{in } |B_2\rangle \end{matrix}$

this measurement can only be <u>measured locally</u>.

If Alice on Earth & Bob on Mars

in a nonlocal condition, you can only do these 2 measurements

$I \otimes Z \rightarrow \zeta_A$

and

$Z \otimes I \rightarrow \zeta_B$

$\boxed{\zeta_A \zeta_B = \begin{matrix} -1 \\ +1 \end{matrix}}$

the correlation allows them to measure the state.

You can always find an observer for which these 2 measurement events are not synchronous.

But just because you couldn't measure the change instantaneously, doesn't mean the change did not happen!

To get $\underline{\zeta_A \zeta_B}$, they have to communicate via signals $\leq$ speed of light

↳ Hence, relativity is not violated.

Holevo Bound:

* Komogorov Complexity of π.

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2} e^{i\varphi}|1\rangle$$

↳ if θ & φ ($\pi/15$) are irrational, it would take an infinite amount of number of bits to precisely specify the state.

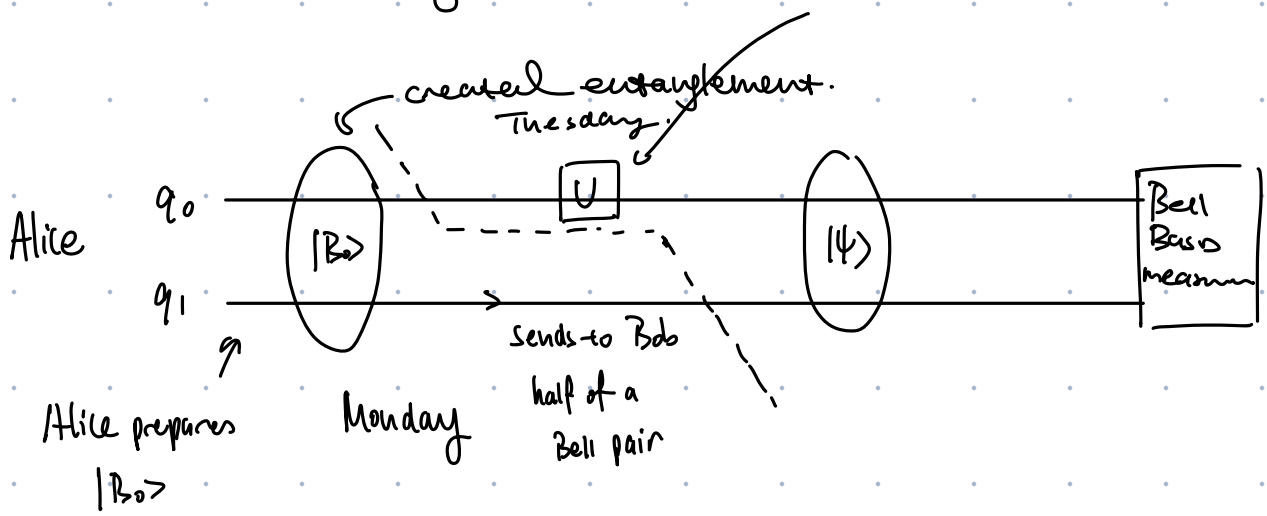
↳ but if you measure the state, you only get 1 bit info.

* ↳ Huge asymmetry betwn # bits to <u>describe the state</u> & <u>measure the state</u>

↳ important for quantum encryption.

Quantum Dense Coding

applies 1 of 4 unitaries U $\begin{cases} I \otimes I \\ I \otimes X \\ I \otimes Y \\ I \otimes Z \end{cases}$



message	U	resulting state $\lvert\psi\rangle$
0 0	$I \otimes Y$	$-i\lvert B_3\rangle$
0 1	$I \otimes X$	$+\lvert B_2\rangle$
1 0	$I \otimes Z$	$-\lvert B_1\rangle$
1 1	$I \otimes I$	$+\lvert B_0\rangle$

→ On Tuesday, after gate operation, Alice sends Bob her qubit & Bob has 2 qubit (2 bits).

→ Alice sent a 1 bit qubit.
Bob now has 2 bits.

How do ppl measure in the Bell basis?

Theorist Method: Construct an operator

$$\hat{M} = \hat{M}^{\dagger} = 0|B_0\rangle\langle B_0| + 1\cdot|B_1\rangle\langle B_1| + 2\cdot|B_2\rangle\langle B_2| + 3\cdot|B_3\rangle\langle B_3|.$$

$$\hat{M}|B_j\rangle = j|B_j\rangle, \quad j = 0,1,2,3.$$

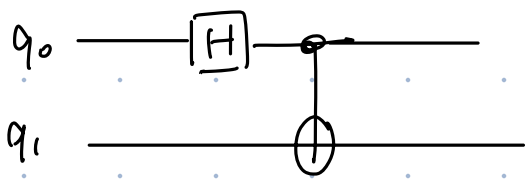
Experimentalist Method: Construct U_D s.t. ← that's so smart!

$$U_D|B_3\rangle = |00\rangle$$

disentangled ⟶ entangled.

$\boxed{H} + \boxed{CNOT}$ → we know.

$$U_E = CNOT_0 (I \otimes H)$$



$U_E|00\rangle = +|B_3\rangle$

$U_E|01\rangle = +|B_2\rangle$

$U_E|10\rangle = +|B_1\rangle$

$U_E|11\rangle = -|B_0\rangle$

← how to run it backwards?

e.g.

$\frac{1}{\sqrt{2}}[|00\rangle + |11\rangle]$

↓ CNOT

$\frac{1}{\sqrt{2}}[|00\rangle + |01\rangle]$

↓

$|0\rangle \otimes \frac{1}{\sqrt{2}}[|0\rangle + |1\rangle]$

↓ H on q_0

$|0\rangle \otimes |0\rangle$

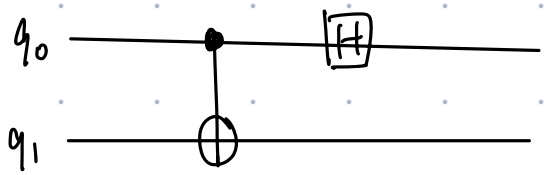
!

inverse of U_E

$$U_D = U_E^\dagger = (I \otimes H)^\dagger (CNOT_0)^\dagger = (I \otimes H)(CNOT_0)$$

why?

U_D



$\underbrace{(CNOT_0)(CNOT_0)}_{\text{inverse}} = I.$

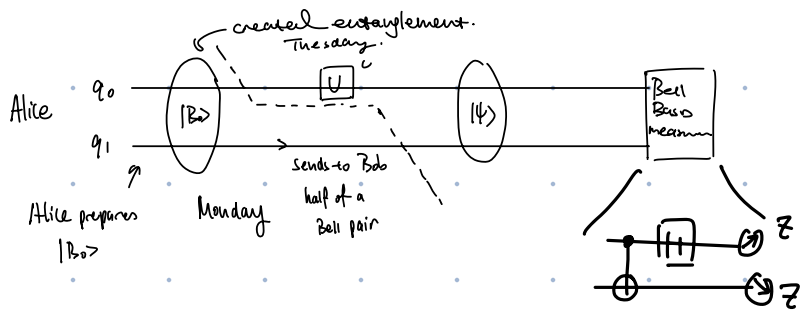
$\underbrace{HH}_{\text{inverse}} = I$

$$U_D|B_3\rangle = |00\rangle$$

$$U_D|B_2\rangle = |01\rangle$$

$$U_D|B_1\rangle = |10\rangle$$

$$U_D|B_0\rangle = -|11\rangle$$



message	U	resulting state $\|\psi\rangle$	$U_D\|\psi\rangle$	Bob's Measurement
00	$I \otimes Y$	$-i\|B_3\rangle$	$-i\|00\rangle$	00, P=1
01	$I \otimes X$	$+\|B_2\rangle$	$+\|01\rangle$	01, P=1
10	$I \otimes Z$	$-\|B_1\rangle$	$-\|10\rangle$	10, P=1
11	$I \otimes I$	$+\|B_0\rangle$	$-\|11\rangle$	11, P=1

Max. amount of qubits

↳ We don't have a complete understanding of diff. classes of entanglement.

↳ $|GHZ\rangle = \frac{1}{\sqrt{2}}\{|000\cdots\cdots0\rangle + |111\cdots\cdots1\rangle\}$.